

Homework 1 Mathematical Foundations of Reinforcement Learning

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Question 1

Part a)

The period of a state i in a Markov chain is defined as

$$d(i) = \gcd\{n \geq 1 : P_{ii}^n > 0\},$$

where P_{ii}^n is the probability of returning to state i after n steps.

If there exists a state j in the state space X with a self-loop, such that $P_{jj} > 0$, then $d(j) = 1$, since it is possible to return to j after one step.

In an irreducible chain, there exists a path between any two states i and j . Thus, there exists a path from i to j and a path from j back to i . Therefore, any return path from j to j can be combined with these paths to create a return path from i to i .

Assume that a return path from i to i has length n_i , where $n_i \bmod d(i) = 0$. Due to irreducibility, this path can pass through state j and be extended by a return path from j to j of length n_j . The resulting path from i to i has length $n_i + n_j$. Therefore,

$$n_i + n_j \equiv 0 \pmod{d(i)}.$$

Since $n_i \equiv 0 \pmod{d(i)}$, it follows that

$$n_j \equiv 0 \pmod{d(i)}.$$

Thus, the return times of j must also be divisible by $d(i)$, which means that $d(i)$ divides $d(j)$ and vice versa. Therefore,

$$d(i) = d(j), \quad \forall i, j \in X.$$

Since $d(j) = 1$, it follows that

$$d(i) = 1 \quad \forall i \in X.$$

Thus, if an irreducible Markov chain contains a self-loop, then the entire chain is aperiodic.

Part b)

A simple example of an irreducible and aperiodic Markov chain is a chain where the possible return paths to a state have lengths 2 and 3. Since

$$\gcd(2, 3) = 1,$$

the chain is aperiodic.

Consider a Markov chain with three states, where from state 1 we can move to state 2, and from state 2 we can either return to state 1 or move to state 3. From state 3, we can then return to state 1. For example, the transition matrix could be

$$P = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{3} & 0 & \frac{2}{3} \\ 1 & 0 & 0 \end{bmatrix}$$

In this chain, there is a path from state 1 back to itself in 2 steps,

$$1 \rightarrow 2 \rightarrow 1,$$

and a path from state 1 back to itself in 3 steps,

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 1.$$

Therefore, the period of state 1 is

$$d(1) = \gcd(2, 3) = 1.$$

Since the chain is irreducible, all states have the same period. Hence,

$$d(i) = 1, \quad \forall i \in X,$$

and the Markov chain is therefore aperiodic.

Question 2

Part a)

Pick an initial reference state x_0 such that $x_0 \in C_0$ and create a general definition of C_i

$$C_i = \{x : \text{there is a path from } x_0 \text{ to } x \text{ whose length is } i \pmod{b}\}.$$

We first check that this is well-defined. Suppose there are two paths from x_0 to the same state x , with lengths m and n . Since the chain is irreducible, there exist a path from x back to x_0 , of some length r . Then $m + r$ and $n + r$ are both return times to x_0 .

Because the period is b , every return time is divisible by b . Hence

$$b \mid (m + r), \quad b \mid (n + r),$$

so

$$b \mid (m - n).$$

Thus

$$m \equiv n \pmod{b}.$$

Therefore each state belongs to exactly one C_i , so X is partitioned into

$$C_1, \dots, C_b.$$

Finally, if $x \in C_i$ and $P(x, y) > 0$, then one extra step takes us from x_0 to y . Hence the path length increases by 1, so

$$y \in C_{i+1},$$

where the index is taken modulo b .

Therefore,

$$P(x, y) > 0$$

only if $x \in C_i$ and $y \in C_{i+1}$ for some i .

Part b)

For each row i in the transition P matrix, there will be a set of N_i elements that can be non-zero, with the rest being zero, where non-zero transition probabilities correspond to the entries in the next set in the cycle we described above C_{i+1} (with modulu). Since there is no overlap between any C_i the rows will have a cyclic block structure.

Question 3

Part a

Our goal is to prove that if P is a symmetric transition matrix and π is the uniform distribution, then π is the stationary distribution P (i.e., $\pi = \pi P$).

We start by observing a unique attribute of a symmetric transition matrix. A transition matrix P is row-stochastic, thus each row $P(i, :)$ always sums to 1. Then if P is also symmetric, each column $P(:, j)$ also sums to 1, since

$$\sum_{i=1}^N p_{ij} = \sum_{i=1}^N p_{ji} = \sum_{i=1}^N p_{ji} = 1.$$

We then look at the definition of the stationary distribution. Let π be the uniform distribution over a state space of size N . Then

$$\pi_i = \frac{1}{N} \quad \forall i \in X.$$

The probability of being in state j after one transition is given by the j -th component of πP . Thus,

$$(\pi P)_j = \sum_{i=1}^N \pi_i p_{ij}.$$

Since π is uniform, we have

$$(\pi P)_j = \sum_{i=1}^N \frac{1}{N} p_{ij} = \frac{1}{N} \sum_{i=1}^N p_{ij}.$$

Because P is symmetric, its columns also sum to 1. Thus,

$$(\pi P)_j = \frac{1}{N} \cdot 1 = \frac{1}{N} = \pi_j.$$

Since this holds for every state j , we can conclude that

$$\pi P = \pi.$$

Thus, the uniform distribution is a stationary distribution of a symmetric transition matrix.

Part b)

As shown above, the only requirement for P to have the uniform distribution as its stationary distribution is that each column sums to 1. It is trivial to create such a matrix. For example,

$$P = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{2}{3} & 0 & \frac{1}{3} \end{bmatrix}$$

Question 4

Part a)

We know that the state distribution at time t in any markov chain can be expressed as

$$\Pr(X_t = x \mid X_0 = x_0) = P^t(x_0, x),$$

We can then replace the expected reward part of the value definition:

$$\begin{aligned} V_T(x_0) &= \frac{1}{T+1} \mathbf{E} \left[\sum_{t=0}^T R(X_t) \mid X_0 = x_0 \right] \\ &= \frac{1}{T+1} \sum_{t=0}^T \sum_{x \in X} P^t(x_0, x) R(x) \\ &= \sum_{x \in X} \left(\frac{1}{T+1} \sum_{t=0}^T P^t(x_0, x) \right) R(x). \end{aligned}$$

Since the chain is irreducible and aperiodic, we know the chain has a unique stationary distribution its state distribution will converge to the stationary distribution

$$\lim_{t \rightarrow \infty} P^t(x_0, x) = \pi(x).$$

Cesaro's theorem says that if some sequence converges to a some value L , then the sequence of its arithmetic means also converges to L , meaning that expression above also converges to the stationary distribution

$$\lim_{T \rightarrow \infty} \frac{1}{T+1} \sum_{t=0}^T P^t(x_0, x) = \pi(x).$$

Therefore,

$$\begin{aligned}
V(x_0) &= \lim_{T \rightarrow \infty} V_T(x_0) \\
&= \lim_{T \rightarrow \infty} \sum_{x \in X} \left(\frac{1}{T+1} \sum_{t=0}^T P^t(x_0, x) \right) R(x) \\
&= \sum_{x \in X} \pi(x) R(x).
\end{aligned}$$

Part b)

Let

$$Y_n = X_{n, \tau_{\text{mix}}(\epsilon)}$$

and

$$\widehat{V}_N = \frac{1}{N} \sum_{n=1}^N R(Y_n).$$

We also have

$$V = \sum_{x \in \mathcal{X}} \pi(x) R(x).$$

First, we use bias–variance decomposition to simplify the expression. For any random variable Z and constant a ,

$$\mathbf{E}[(Z - a)^2] = \text{Var}(Z) + (\mathbf{E}[Z] - a)^2.$$

Hence, we divide the expression into two separate components which we will find the upper bounds for

$$\mathbf{E}[(\widehat{V}_N - V)^2] = \text{Var}(\widehat{V}_N) + (\mathbf{E}[\widehat{V}_N] - V)^2.$$

We start by finding the upper bound for $\text{Var}(\widehat{V}_N)$. Since the N simulations are independent, then

$$\text{Var}(\widehat{V}_N) = \frac{1}{N^2} \sum_{n=1}^N \text{Var}(R(Y_n)).$$

We know from the question that

$$-R_{\max} \leq R(Y_n) \leq R_{\max},$$

Popoviciu's inequality then gives

$$\text{Var}(R(Y_n)) \leq \frac{(R_{\max} - (-R_{\max}))^2}{4} = \frac{4R_{\max}^2}{4} = R_{\max}^2.$$

Therefore, the upper bound is

$$\text{Var}(\widehat{V}_N) \leq \frac{R_{\max}^2}{N}.$$

Now let

$$p_\epsilon(x) = \mathbf{P}(Y_n = x).$$

Then

$$\mathbf{E}[\widehat{V}_N] = \sum_{x \in \mathcal{X}} p_\epsilon(x) R(x)$$

We then compute an upper bound of $(\mathbf{E}[\widehat{V}_N] - V)^2$. Recall we previously showed that $V = \sum_{x \in \mathcal{X}} \pi(x) R(x)$ in a). Combining that, the triangle inequality, and the fact that we know the bound of R is $R_{\max}$ we can then express upper bound as

$$\begin{aligned} \left| \mathbf{E}[\widehat{V}_N] - V \right| &= \left| \sum_{x \in \mathcal{X}} (p_\epsilon(x) - \pi(x)) R(x) \right| \\ &\leq \sum_{x \in \mathcal{X}} |p_\epsilon(x) - \pi(x)| |R(x)| \\ &\leq R_{\max} \sum_{x \in \mathcal{X}} |p_\epsilon(x) - \pi(x)|. \end{aligned}$$

By the definition of the mixing time, we know that the ϵ bounds the error as

$$\frac{1}{2} \sum_{x \in \mathcal{X}} |p_\epsilon(x) - \pi(x)| \leq \epsilon,$$

so

$$\sum_{x \in \mathcal{X}} |p_\epsilon(x) - \pi(x)| \leq 2\epsilon.$$

Thus,

$$\left| \mathbf{E}[\widehat{V}_N] - V \right| \leq 2R_{\max}\epsilon,$$

and hence

$$(\mathbf{E}[\widehat{V}_N] - V)^2 \leq 4R_{\max}^2\epsilon^2.$$

Combining the two bounds gives us the desired expression

$$\begin{aligned} \mathbf{E}[(\widehat{V}_N - V)^2] &\leq \frac{R_{\max}^2}{N} + 4R_{\max}^2\epsilon \\ &= R_{\max}^2 \left(\frac{1}{N} + 4\epsilon^2 \right). \end{aligned}$$

Part c)

In this part, ϵ denotes the desired accuracy of the estimator with respect to the MSE, while η is the mixing accuracy from now on. We want

$$\mathbf{E}[(\widehat{V}_N - V)^2] \leq \epsilon.$$

From part (b), if each chain is run for $\tau_{\text{mix}}(\eta)$ steps, then

$$\mathbf{E}[(\widehat{V}_N - V)^2] \leq R_{\text{max}}^2 \left(\frac{1}{N} + 4\eta^2 \right),$$

where η is the corresponding mixing accuracy.

We choose

$$N \geq \frac{2R_{\text{max}}^2}{\epsilon}$$

and run each chain for

$$\tau_{\text{mix}} \left(\sqrt{\frac{\epsilon}{8R_{\text{max}}^2}} \right)$$

steps.

With this choice,

$$\frac{R_{\text{max}}^2}{N} \leq \frac{\epsilon}{2},$$

and

$$4R_{\text{max}}^2 \left(\sqrt{\frac{\epsilon}{8R_{\text{max}}^2}} \right)^2 = \frac{\epsilon}{2}.$$

Therefore,

$$\mathbf{E}[(\widehat{V}_N - V)^2] \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

It remains to bound the number of steps needed for each simulation. Using the mixing time bound in the hint

$$\tau_{\text{mix}}(\eta) \leq \lceil \log_2(\eta^{-1}) \rceil \tau_{\text{mix}}(1/4),$$

with

$$\eta = \sqrt{\frac{\epsilon}{8R_{\max}^2}},$$

we get

$$\tau_{\text{mix}}\left(\sqrt{\frac{\epsilon}{8R_{\max}^2}}\right) \leq \left\lceil \log_2\left(\sqrt{\frac{8R_{\max}^2}{\epsilon}}\right) \right\rceil \tau_{\text{mix}}(1/4).$$

Since

$$\log_2\left(\sqrt{\frac{8R_{\max}^2}{\epsilon}}\right) = \frac{1}{2} \log_2\left(\frac{8R_{\max}^2}{\epsilon}\right),$$

the length of each simulation can be bounded by

$$\left\lceil \frac{1}{2} \log_2\left(\frac{8R_{\max}^2}{\epsilon}\right) \right\rceil \tau_{\text{mix}}(1/4).$$

Thus, an ϵ -accurate estimator can be obtained using

$$N = \left\lceil \frac{2R_{\max}^2}{\epsilon} \right\rceil$$

independent simulations, each of length at most

$$\left\lceil \frac{1}{2} \log_2\left(\frac{8R_{\max}^2}{\epsilon}\right) \right\rceil \tau_{\text{mix}}(1/4).$$

Question 5

Part a)

Let $x^*, y^* \in X$ both be fixed points of T . Then

$$T(x^*) = x^* \quad \text{and} \quad T(y^*) = y^*.$$

Since T is an α -contraction,

$$\|x^* - y^*\| = \|T(x^*) - T(y^*)\| \leq \alpha \|x^* - y^*\|,$$

where $0 \leq \alpha < 1$. Therefore,

$$(1 - \alpha)\|x^* - y^*\| \leq 0.$$

Since $1 - \alpha > 0$ and $\|x^* - y^*\| \geq 0$, it follows that

$$\|x^* - y^*\| = 0.$$

By the positive definiteness property of a norm,

$$\|x^* - y^*\| = 0 \implies x^* - y^* = 0.$$

Hence,

$$x^* = y^*.$$

Therefore, if T has a fixed point in X , it is unique.

Part b

To construct an α -contraction with no fixed point, we want the steps of T to approach a point that does not belong to the support X . This can happen when X is not complete.

For example, say

$$X = (0, 1)$$

with the usual norm $\|x\| = |x|$, and define a function

$$T(x) = \frac{x + 1}{2}.$$

For any $x, y \in X$,

$$\|T(x) - T(y)\| = \left| \frac{x + 1}{2} - \frac{y + 1}{2} \right| = \frac{1}{2}|x - y|.$$

meaning T is an α -contraction with

$$\alpha = \frac{1}{2}.$$

A fixed point x^* would have to satisfy

$$T(x^*) = x^*.$$

Thus,

$$\frac{x^* + 1}{2} = x^*,$$

which implies

$$x^* = 1.$$

However,

$$1 \notin X = (0, 1).$$

Therefore, 1 is not a fixed point. Furthermore, we know from the previous question that if X was complete, 1 would be the one unique fixed point. Therefore T has no fixed point in X .

Part c)

We want to show that repeated application of the contraction produces a Cauchy sequence. Since X is complete, this sequence must converge to some point $x^* \in X$. We then show that this limit point is a fixed point of T .

Choose some start value $x_0 \in X$, and define the sequence

$$x_{n+1} = T(x_n).$$

Since T is an α -contraction,

$$\|x_{n+1} - x_n\| = \|T(x_n) - T(x_{n-1})\| \leq \alpha \|x_n - x_{n-1}\|.$$

Applying this repeatedly gives

$$\|x_{n+1} - x_n\| \leq \alpha^n \|x_1 - x_0\|.$$

Now let $m > n$. By the triangle inequality,

$$\|x_m - x_n\| \leq \|x_m - x_{m-1}\| + \cdots + \|x_{n+1} - x_n\|.$$

Using the bound above,

$$\|x_m - x_n\| \leq (\alpha^{m-1} + \cdots + \alpha^n) \|x_1 - x_0\|.$$

Since $0 \leq \alpha < 1$, by the geometric series formula we have

$$\alpha^n + \alpha^{n+1} + \cdots = \frac{\alpha^n}{1 - \alpha},$$

and therefore

$$\|x_m - x_n\| \leq \frac{\alpha^n}{1 - \alpha} \|x_1 - x_0\|.$$

Because $\alpha^n \rightarrow 0$, for every $\varepsilon > 0$ there exists N such that

$$m, n \geq N \implies \|x_m - x_n\| < \varepsilon.$$

Hence $x_0, x_1, \dots$ is a Cauchy sequence.

Since X is complete, there exists some $x^* \in X$ such that

$$x_n \rightarrow x^*.$$

We need also to show that x^* is a fixed point. Since

$$x_{n+1} = T(x_n),$$

and $x_n \rightarrow x^*$, we also have

$$x_{n+1} \rightarrow x^*.$$

Furthermore, by the contraction property,

$$\|T(x_n) - T(x^*)\| \leq \alpha \|x_n - x^*\|.$$

Since $x_n \rightarrow x^*$, the right-hand side tends to 0. Thus

$$T(x_n) \rightarrow T(x^*).$$

But $T(x_n) = x_{n+1}$, and we already know that

$$x_{n+1} \rightarrow x^*.$$

Therefore the sequence $T(x_n)$ converges both to $T(x^*)$ and to x^* . Since limits are unique,

$$T(x^*) = x^*.$$

Thus T has a fixed point in X .

Question 6

Part a)

We show that the mapping

$$T(x) = Wx$$

is not contractive with respect to any norm. We will prove this by contradiction.

W is doubly stochastic. Every column and every row of W sums to one. Hence,

$$W\mathbf{1} = \mathbf{1},$$

where $\mathbf{1} \in \mathbf{R}^N$ denotes the vector of all ones. Thus, $\mathbf{1}$ is an eigenvector of W corresponding to the eigenvalue 1.

Suppose, for contradiction, that $T(x) = Wx$ is contractive in some norm $\|\cdot\|$. Then there exists a constant $c < 1$ such that

$$\|Wx - Wy\| \leq c\|x - y\|$$

for all $x, y \in \mathbf{R}^N$.

Choosing $x = \mathbf{1}$ and $y = 0$, we obtain

$$\|W\mathbf{1} - W0\| \leq c\|\mathbf{1} - 0\|.$$

Since $W\mathbf{1} = \mathbf{1}$ and $W0 = 0$, this becomes

$$\|\mathbf{1}\| \leq c\|\mathbf{1}\|.$$

Because $\mathbf{1} \neq 0$, we have $\|\mathbf{1}\| > 0$, and therefore

$$1 \leq c,$$

which contradicts the assumption that $c < 1$.

Hence, the mapping $x \mapsto Wx$ is not contractive in any norm.

Part b)

We show that the set of fixed points of the algorithm is

$$\text{span}(\mathbf{1}) = \{a\mathbf{1} \in \mathbf{R}^N : a \in \mathbf{R}\}.$$

A vector $x \in \mathbf{R}^N$ is a fixed point of the algorithm if

$$Wx = x.$$

Thus, the fixed points are exactly the eigenvectors of W corresponding to the eigenvalue 1.

Since W is doubly stochastic, each row sums to one. Therefore,

$$W\mathbf{1} = \mathbf{1},$$

which shows that 1 is an eigenvalue of W , with corresponding eigenvector $\mathbf{1}$.

We are given that W is irreducible, non-negative, and has spectral radius

$$\rho(W) = 1.$$

By the Perron–Frobenius theorem provided in the question, the eigenvalue equal to the spectral radius is simple. So, the eigenspace corresponding to the eigenvalue 1 is one-dimensional.

Since $\mathbf{1}$ is an eigenvector corresponding to 1, this eigenspace must be

$$\text{span}(\mathbf{1}).$$

Therefore, the set of fixed points of the algorithm is all scalar multiples of the $\mathbf{1}$ vector

$$\text{span}(\mathbf{1}) = \{a\mathbf{1} : a \in \mathbf{R}\}.$$

Part c

Recall that

$$A(\bar{x}) = \left\{ x \in \mathbf{R}^N : \frac{1}{N} \sum_{n=1}^N x_n = \bar{x} \right\}.$$

Let $x \in A(\bar{x})$. Then

$$\frac{1}{N} \mathbf{1}^T x = \bar{x}.$$

We want to show that $Wx \in A(\bar{x})$. Since W is doubly stochastic, its columns also sum to one, and therefore

$$\mathbf{1}^T W = \mathbf{1}^T.$$

Hence,

$$\frac{1}{N} \mathbf{1}^T Wx = \frac{1}{N} \mathbf{1}^T x = \bar{x}.$$

Thus Wx has the same average as x , so

$$Wx \in A(\bar{x}).$$

Therefore,

$$x \in A(\bar{x}) \implies Wx \in A(\bar{x}).$$

Part d

Let $x \in A(\bar{x})$. Since

$$\frac{1}{N} \mathbf{1}^T x = \bar{x},$$

we have

$$\frac{1}{N} \mathbf{1} \mathbf{1}^T x = \bar{x} \mathbf{1}.$$

Therefore,

$$\begin{aligned} Wx - \bar{x} \mathbf{1} &= Wx - \frac{1}{N} \mathbf{1} \mathbf{1}^T x \\ &= \left(W - \frac{1}{N} \mathbf{1} \mathbf{1}^T \right) x. \end{aligned}$$

Furthermore,

$$\left(W - \frac{1}{N} \mathbf{1} \mathbf{1}^T \right) \mathbf{1} = W\mathbf{1} - \frac{1}{N} \mathbf{1} \mathbf{1}^T \mathbf{1} = \mathbf{1} - \mathbf{1} = \mathbf{0}.$$

Hence,

$$Wx - \bar{x}\mathbf{1} = \left(W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right)(x - \bar{x}\mathbf{1}).$$

Using the induced Euclidean norm, we obtain

$$\|Wx - \bar{x}\mathbf{1}\|_2 \leq \left\|W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right\|_2 \|x - \bar{x}\mathbf{1}\|_2.$$

Define

$$\sigma = \left\|W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right\|_2.$$

Thus,

$$\|Wx - \bar{x}\mathbf{1}\|_2 \leq \sigma \|x - \bar{x}\mathbf{1}\|_2.$$

It remains to show that $\sigma < 1$. Since W is doubly stochastic,

$$W\mathbf{1} = \mathbf{1}, \quad W^T\mathbf{1} = \mathbf{1}.$$

Thus, 1 is the largest singular value of W . Moreover, W^TW is a non-negative doubly stochastic matrix. Because the communication graph is connected and $W_{ii} > 0$, the matrix W^TW is irreducible.

By the Perron–Frobenius theorem, the eigenvalue 1 of W^TW is simple. The corresponding eigenvector is $\mathbf{1}$. Therefore, all other eigenvalues of W^TW are strictly smaller than 1.

The matrix

$$\frac{1}{N}\mathbf{1}\mathbf{1}^T$$

removes the component in the direction of $\mathbf{1}$. Hence,

$$\sigma = \left\|W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right\|_2 < 1.$$

Consequently,

$$\|Wx - \bar{x}\mathbf{1}\|_2 \leq \sigma \|x - \bar{x}\mathbf{1}\|_2, \quad \sigma < 1.$$

Thus, W is a pseudo-contraction on the set $A(\bar{x})$, since every iteration strictly contracts the distance to the fixed point $\bar{x}\mathbf{1}$.

Part e

Since W is symmetric, it has an orthonormal basis of eigenvectors $q_1, \dots, q_N$ with corresponding real eigenvalues

$$\lambda_1, \lambda_2, \dots, \lambda_N.$$

Because W is doubly stochastic,

$$W\mathbf{1} = \mathbf{1},$$

so 1 is an eigenvalue of W . We may therefore choose

$$q_1 = \frac{1}{\sqrt{N}}\mathbf{1}, \quad \lambda_1 = 1.$$

By the Perron–Frobenius theorem, since W is non-negative and irreducible, the eigenvalue 1 is simple. Moreover, because $W_{ii} > 0$, all other eigenvalues satisfy

$$|\lambda_i| < 1, \quad i = 2, \dots, N.$$

Let

$$|\lambda_2| = \max_{i \geq 2} |\lambda_i|.$$

Since the eigenvectors form an orthonormal basis, we can write the initial vector as

$$x(0) = \sum_{i=1}^N \alpha_i q_i.$$

The coefficient corresponding to q_1 is

$$\alpha_1 = q_1^T x(0) = \frac{1}{\sqrt{N}} \mathbf{1}^T x(0).$$

Since

$$\bar{x} = \frac{1}{N} \mathbf{1}^T x(0),$$

we obtain

$$\alpha_1 = \sqrt{N}\bar{x}.$$

Therefore,

$$\alpha_1 q_1 = \sqrt{N}\bar{x} \frac{1}{\sqrt{N}} \mathbf{1} = \bar{x} \mathbf{1}.$$

Thus,

$$x(0) = \bar{x} \mathbf{1} + \sum_{i=2}^N \alpha_i q_i.$$

After k iterations,

$$x(k) = W^k x(0) = \bar{x} \mathbf{1} + \sum_{i=2}^N \alpha_i \lambda_i^k q_i.$$

Hence,

$$x(k) - \bar{x} \mathbf{1} = \sum_{i=2}^N \alpha_i \lambda_i^k q_i.$$

Using the orthonormality of the eigenvectors,

$$\begin{aligned} \|x(k) - \bar{x} \mathbf{1}\|_2^2 &= \sum_{i=2}^N \alpha_i^2 |\lambda_i|^{2k} \\ &\leq |\lambda_2|^{2k} \sum_{i=2}^N \alpha_i^2. \end{aligned}$$

Taking square roots gives

$$\|x(k) - \bar{x} \mathbf{1}\|_2 \leq |\lambda_2|^k \left(\sum_{i=2}^N \alpha_i^2 \right)^{1/2}.$$

Define

$$A = \left(\sum_{i=2}^N \alpha_i^2 \right)^{1/2} = \|x(0) - \bar{x}\mathbf{1}\|_2.$$

Therefore,

$$\|x(k) - \bar{x}\mathbf{1}\|_2 \leq A|\lambda_2|^k.$$

Since

$$|\lambda_2| < 1,$$

it follows that

$$\|x(k) - \bar{x}\mathbf{1}\|_2 \rightarrow 0$$

as $k \rightarrow \infty$. Hence,

$$x(k) \rightarrow \bar{x}\mathbf{1}.$$

Thus, the algorithm converges linearly to the average consensus vector, with convergence rate determined by the second largest eigenvalue of W in absolute value.

Question 7

We consider the following algorithm

$$x(k+1) = x(k) - \gamma(\nabla f(x(k)) + e(k)).$$

Let

$$d_k = x(k) - x^*,$$

where x^* is the minimizer of f . Since f is differentiable and strongly convex,

$$\nabla f(x^*) = 0.$$

We can then formulate our target expression,

$$d_{k+1} = d_k - \gamma(\nabla f(x(k)) + e(k)).$$

Taking the squared norm, we get

$$\|d_{k+1}\|^2 = \|d_k - \gamma\nabla f(x(k)) - \gamma e(k)\|^2.$$

Our goal is now to identify above expressions expected value's upper bound. Since $e(k)$ is zero mean and independent of $x(k)$, the cross terms involving $e(k)$ vanish after taking expectation. Also,

$$\mathbf{E}\|e(k)\|^2 = \sigma^2.$$

Hence,

$$\begin{aligned}\mathbf{E}\|d_{k+1}\|^2 &= \mathbf{E}\|d_k\|^2 - 2\gamma\mathbf{E}[\langle \nabla f(x(k)), d_k \rangle] \\ &\quad + \gamma^2\mathbf{E}\|\nabla f(x(k))\|^2 + \gamma^2\sigma^2.\end{aligned}$$

Now we use the inequality from the hint:

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \frac{L\mu}{L+\mu}\|x - y\|^2 + \frac{1}{L+\mu}\|\nabla f(x) - \nabla f(y)\|^2.$$

Take $x = x(k)$ and $y = x^*$. Since $\nabla f(x^*) = 0$, this becomes

$$\langle \nabla f(x(k)), d_k \rangle \geq \frac{L\mu}{L+\mu}\|d_k\|^2 + \frac{1}{L+\mu}\|\nabla f(x(k))\|^2.$$

Substituting this into the previous expression gives

$$\begin{aligned}\mathbf{E}\|d_{k+1}\|^2 &\leq \left(1 - \frac{2\gamma\mu L}{\mu + L}\right)\mathbf{E}\|d_k\|^2 \\ &\quad + \left(\gamma^2 - \frac{2\gamma}{\mu + L}\right)\mathbf{E}\|\nabla f(x(k))\|^2 + \gamma^2\sigma^2.\end{aligned}$$

We notice that

$$\gamma^2 - \frac{2\gamma}{\mu + L} = \gamma\left(\gamma - \frac{2}{\mu + L}\right).$$

As per the definition in the question,

$$0 < \gamma < \frac{2}{L + \mu},$$

thus quantity is negative. Therefore, the gradient-norm term is nonpositive and can be dropped from the upper bound. Thus,

$$\mathbf{E}\|d_{k+1}\|^2 \leq \left(1 - \frac{2\gamma\mu L}{\mu + L}\right)\mathbf{E}\|d_k\|^2 + \gamma^2\sigma^2.$$

Define

$$\rho = 1 - \frac{2\gamma\mu L}{\mu + L}.$$

Then

$$\mathbf{E}\|d_{k+1}\|^2 \leq \mathbf{E}\|d_k\|^2 + \gamma^2\sigma^2.$$

Since $0 < \rho < 1$, applying this inequality recursively gives

$$\mathbf{E}\|d_{k+1}\|^2 \leq \rho^{k+1}\mathbf{E}\|d_0\|^2 + \gamma^2\sigma^2 \sum_{j=0}^k \rho^j.$$

Using the geometric series bound

$$\sum_{j=0}^k \rho^j \leq \frac{1}{1-\rho},$$

we obtain

$$\mathbf{E}\|d_{k+1}\|^2 \leq \rho^{k+1}\mathbf{E}\|d_0\|^2 + \frac{\gamma^2\sigma^2}{1-\rho}.$$

Since

$$1 - \rho = \frac{2\gamma\mu L}{\mu + L},$$

we finally get the expression we were looking for

$$\mathbf{E}\|x(k+1) - x^*\|^2 \leq \left(1 - \frac{2\gamma\mu L}{\mu + L}\right)^{k+1} \mathbf{E}\|x(0) - x^*\|^2 + \frac{\gamma\sigma^2(\mu + L)}{2\mu L}.$$

The first term decreases as k increases, so it eventually goes to zero. Therefore,

$$\lim_{k \rightarrow \infty} \mathbf{E}\|x(k) - x^*\|^2 \leq \frac{\gamma\sigma^2(\mu + L)}{2\mu L}.$$

This means that, even when the noise variance σ^2 is positive and large we can make the final expected error arbitrarily small by choosing γ sufficiently small. In fact, the error is proportional to γ .

However, there is an obvious tradeoff. The convergence factor is

$$1 - \frac{2\gamma\mu L}{\mu + L}.$$

and if γ is chosen very small, this factor moves towards 1 and the algorithm converges very slowly. We can therefore conclude that smaller step size gives better final solution accuracy, but the cost is a larger number of iterations needed to reach that accuracy.

Question 8

We start by describing the two convergence types in detail to better understand their differences.

Almost sure convergence means that a sequence of random variables X_n converges to some limit X with probability 1. I.e.,

$$P\left(\lim_{n \rightarrow \infty} X_n = X\right) = 1.$$

A good way to think about this is to fix one particular outcome ω and look at

$$X_1(\omega), X_2(\omega), X_3(\omega), \dots$$

Almost sure convergence says that, for almost every ω , this sequence eventually settles down and converges to $X(\omega)$. There can be some exceptional outcomes where this does not happen, but those outcomes must have probability 0.

Convergence in probability is weaker. It only requires that the probability of being far from the limit goes to zero:

$$P(|X_n - X| \geq \varepsilon) \rightarrow 0.$$

So we do not require every fixed outcome to eventually settle down. We only require that, at step n , the probability of getting a value far from X becomes smaller and smaller.

A sequence that converge in probability but not almost surely is therefore any sequence where the chance of a value far away from the limit has a decreasing but non-zero chance of being realized. A common example for this is

$$X_n \sim \text{Bernoulli}\left(\frac{1}{n}\right),$$

with independent X_n 's. This means

$$X_n = \begin{cases} 1 & \text{with probability } 1/n, \\ 0 & \text{with probability } 1 - 1/n. \end{cases}$$

Take the limit to be $X = 0$. The value 1 can be thought of as the outlier, since it is far from 0. Its probability is

$$P(X_n = 1) = \frac{1}{n} \rightarrow 0.$$

Therefore X_n converges to 0 in probability.

But, this does not mean a fixed realization eventually becomes 0 forever. Since the X_n 's are independent, a realization can keep producing occasional 1's infinitely far into the sequence. In fact, since

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty,$$

the value 1 occurs infinitely often with probability 1. So the sequence does not converge almost surely to 0.

The main difference is therefore that convergence in probability only asks for the probability of an outlier at step n to go to zero, while almost sure convergence asks that, for almost every fixed realization, the values eventually stay close to the limit.

Question 9

Part a

We want to prove Chebyshev's inequality using Markov's inequality.

Markov's inequality states that if Z is a non-negative random variable, then for every $a > 0$,

$$P(Z \geq a) \leq \frac{E[Z]}{a}.$$

Let Y be a random variable with finite mean $\mu = E[Y]$ and finite variance. We want to bound

$$P(|Y - \mu| \geq \alpha).$$

To apply Markov's inequality, we need a non-negative random variable. Therefore, consider

$$Z = (Y - \mu)^2.$$

Since Z is a square, we always have $Z \geq 0$. Also, for $\alpha > 0$,

$$|Y - \mu| \geq \alpha$$

then

$$(Y - \mu)^2 \geq \alpha^2.$$

Thus,

$$P(|Y - \mu| \geq \alpha) = P((Y - \mu)^2 \geq \alpha^2).$$

Now we use Markov's inequality on the non-negative random variable $(Y - \mu)^2$:

$$P((Y - \mu)^2 \geq \alpha^2) \leq \frac{E[(Y - \mu)^2]}{\alpha^2}.$$

By the definition of variance,

$$\text{Var}(Y) = E[(Y - \mu)^2].$$

Therefore,

$$P(|Y - \mu| \geq \alpha) \leq \frac{\text{Var}(Y)}{\alpha^2}.$$

Part b

Let

$$X_1, X_2, \dots, X_n$$

be IID random variables with

$$E[X_i] = \mu$$

and

$$\text{Var}(X_i) = \sigma^2.$$

We define the sample mean as

$$S_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

We want to use Chebyshev's inequality to show that S_n converges in probability to μ .

First, we calculate the expected value of S_n . Using linearity of expectation,

$$E[S_n] = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right].$$

We can move the constant $1/n$ outside the expectation and split the sum:

$$E[S_n] = \frac{1}{n} \sum_{i=1}^n E[X_i].$$

Since all the X_i 's have mean μ ,

$$E[S_n] = \frac{1}{n} \sum_{i=1}^n \mu = \frac{1}{n}(n\mu) = \mu.$$

Therefore, the sample mean S_n also has mean μ .

Next, we calculate the variance of S_n :

$$\text{Var}(S_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right).$$

Using the rule

$$\text{Var}(aX) = a^2 \text{Var}(X),$$

we get

$$\text{Var}(S_n) = \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right).$$

Because the random variables $X_1, \dots, X_n$ are independent, the variance of their sum is equal to the sum of their variances. Hence,

$$\text{Var}(S_n) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i).$$

Since each X_i has variance σ^2 ,

$$\text{Var}(S_n) = \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n}.$$

Now we use Chebyshev's inequality to S_n . Since $E[S_n] = \mu$, Chebyshev's inequality gives

$$P(|S_n - \mu| \geq \epsilon) \leq \frac{\text{Var}(S_n)}{\epsilon^2}.$$

Substituting

$$\text{Var}(S_n) = \frac{\sigma^2}{n},$$

we obtain

$$P(|S_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2}.$$

Now let $n \rightarrow \infty$. Since σ^2 and $\epsilon > 0$ are fixed,

$$\frac{\sigma^2}{n\epsilon^2} \rightarrow 0.$$

Therefore,

$$P(|S_n - \mu| \geq \epsilon) \rightarrow 0.$$

This means that for every $\epsilon > 0$, the probability that S_n differs from μ by at least ϵ approaches zero as the number of samples increases.

By the definition of convergence in probability, this shows that

$$S_n \xrightarrow{P} \mu.$$

Thus, the sample mean converges in probability to the true mean μ , which proves the weak law of large numbers.

Part c

We want to determine how large n should be so that the sample mean S_n is within ϵ of the true mean μ with probability at least $1 - \delta$.

In other words, we want

$$P(|S_n - \mu| \leq \epsilon) \geq 1 - \delta.$$

It is often easier to work with the complementary event. The condition above is equivalent to requiring

$$P(|S_n - \mu| \geq \epsilon) \leq \delta.$$

From part (b), Chebyshev's inequality gives

$$P(|S_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2}.$$

Therefore, it is enough to choose n such that

$$\frac{\sigma^2}{n\epsilon^2} \leq \delta.$$

Now we solve this inequality for n . Multiplying both sides by $n\epsilon^2$, we get

$$\sigma^2 \leq n\epsilon^2\delta.$$

Dividing by $\epsilon^2\delta$, we obtain

$$n \geq \frac{\sigma^2}{\epsilon^2\delta}.$$

Since n represents the number of samples, it must be an integer. Therefore, we can choose

$$n \geq \left\lceil \frac{\sigma^2}{\epsilon^2\delta} \right\rceil.$$

With this choice of n , we have

$$P(|S_n - \mu| \leq \epsilon) \geq 1 - \delta.$$

Thus, collecting at least

$$\left\lceil \frac{\sigma^2}{\epsilon^2\delta} \right\rceil$$

samples guarantees, using Chebyshev's inequality, that S_n is an ϵ -approximation of μ with probability at least $1 - \delta$.